

2024



Alphalion Technology Delivers Next-Gen Post-Trade Processing with SaaS Platform Running on AWS

Alphalion Technology's SaaS platform, Falcon, meets growing demand for rapid post-trade processing using Amazon EC2 instances for real-time performance and AWS CloudFormation for simplified provisioning.

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100%

increase in annual revenue growth

70%

Reliably high performance



Continual improvement

with support for new AI use cases

Overview

Fintech [Alphalion Technology](#) (Alphalion) sought a reliable and secure cloud solution for its next-gen post-trade processing SaaS platform, Falcon. It chose Amazon Web Services (AWS), hosting Falcon's applications on [Amazon Elastic Compute Cloud \(Amazon EC2\)](#) instances and using [Amazon Relational Database Service \(Amazon RDS\)](#) for PostgreSQL for its application databases.

By building Falcon on AWS, Alphalion meets the growing demand for SaaS-based real-time post-trade processing. With revenues doubling year-on-year, the company benefits from increased time for innovation facilitated by [AWS CloudFormation](#), which streamlines provisioning.



Opportunity | Running an Automated SaaS Platform on a Secure, Scalable Cloud Infrastructure

[Alphalion Technology](#) (Alphalion) is a fintech startup that offers Falcon, a highly automated SaaS platform for real-time processing of financial trades in middle and back-office operations. Falcon supports a wide range of financial products, including futures, options, forex trading, equities, contracts for difference, fixed income, and others.



The platform simplifies multi-asset post-trade processing for sell-side businesses, such as brokers, banks, and wealth management. These entities increasingly invest in SaaS platforms like Falcon to replace aging systems that lack performance and complicate compliance reporting. Additionally, mid-sized online brokers and newer entrants leverage Falcon to enhance post-trade processing speed in a competitive marketplace.

To meet market demands, Alphalion chose to provide Falcon as a cloud-native solution, prioritizing scalability, reliability, and security to bolster its SaaS platform. In pursuit of a global cloud infrastructure compliant with data localization regulations, Alphalion evaluated multiple cloud providers, ultimately selecting Amazon Web Services (AWS). Lou Wang, CEO, and founder of Alphalion Technology, says, "Trust is crucial in financial services. We wanted a cloud provider that both we and our customers could trust, and AWS stood out."

Solution | Streamlining Deployment and Platform Management on AWS

Alphalion collaborated with its local AWS account team to design Falcon's flexible cloud infrastructure. The platform's cloud-native Linux-based applications, running in Docker containers, is hosted on [Amazon Elastic Compute Cloud \(Amazon EC2\)](#) virtual services. Alphalion also adopted [Amazon Relational Database Service \(Amazon RDS\) for PostgreSQL](#) for its application databases and uses [Amazon Simple Storage Service \(Amazon S3\)](#) to store compliance reports and statements and system backups.

To maximize security, Alphalion utilizes [Amazon Virtual Private Cloud \(Amazon VPC\)](#) to create security groups and network access control lists to govern access to Amazon EC2 instances. Furthermore, using [AWS Direct Connect](#), Alphalion secures data as it moves between customer networks and AWS. The company also adopted [AWS Identity and Access Management \(IAM\)](#) to lock down identities and access AWS services and resources.

In 2023, Alphalion enhanced its application and data security by implementing [AWS Key Management Service \(AWS KMS\)](#) for creating, managing, and controlling cryptographic keys across AWS environments. "Implementing the AWS KMS API [application programming interface] through the AWS SDK was seamless and saved months of development work," says Wang. "AWS KMS provided a secure and scalable encryption and key management solution."



With its AWS environment in place, Alphalion can now focus on platform development rather than infrastructure management. The company utilizes [AWS CloudFormation](#) to deploy security appliances, ensuring consistent environments with each deployment. This approach minimizes time spent managing resources, allowing a greater focus on security applications running on AWS. Wang says, "AWS CloudFormation ensures that each new environment is identical, making it easier to provision so we can focus 70 percent more time on platform development."

Outcome | Supporting 100% Annual Revenue Growth with Falcon on AWS

Running Falcon on AWS has facilitated Alphalion's rapid business expansion, resulting in revenues doubling year-on-year. The global presence of sell-side businesses adopting Falcon as their core platform is continuously growing. "We now have footprints in Singapore, the US, Thailand, Malaysia, and Hong Kong." Wang explains.

With a growing reputation for performance and reliability on AWS, Falcon is becoming the preferred choice across the financial services industry. "Customers often highlight the scalability and security of the system, coupled with the strong support from our development team, as key factors contributing to the success of their businesses," says Wang.

Looking ahead, Alphalion aims to bolster Falcon's capabilities with AI-powered analytics. "We envision numerous AI use cases for Falcon, including anomaly detection, collateral optimization, as well as activity analysis for regulatory and risk purposes," Wang explains. He concludes, "I'm confident that AWS can support us on our AI strategy over the coming years as we bring the benefits of the technology to our customers."

About Alphalion Technology

Headquartered in Hong Kong, Alphalion Technology is an innovator and disruptor in the financial services industry. The company's Falcon SaaS platform serves as the core post-trade transaction processing system infrastructure for clients around the globe.

AWS Services Used

Amazon EC2

Amazon Elastic Compute Cloud (Amazon EC2) offers the broadest and deepest compute platform, with over 750 instances and choice of the latest processor, storage, networking, operating system, and purchase model to help you best match the needs of your workload.

[Learn more »](#)



Amazon RDS for PostgreSQL

[PostgreSQL](#) has become the preferred [open source](#) relational database for many enterprise developers and startups, powering leading business and mobile applications.

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Amazon S3

Amazon Simple Storage Service (Amazon S3) is an object storage service offering industry-leading scalability, data availability, security, and performance.

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AWS Key Management Service

AWS Key Management Service (AWS KMS) lets you create, manage, and control cryptographic keys across your applications and AWS services.

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